# Project 5 — Looking Ahead

**Using data to make predictions.**

# Project 5: Team prediction challenge

**Total points:** 35

In this final project, you will work in a team to use tools from STAT 109 to collect data and make a prediction for your choice of two problems. Each team then submits an **one-page summary** that includes a clear section describing team member contributions.

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## Timeline

- **Wednesday, May 6 (class):** choose team, choose project option, finalize variables, collect data.
- **Thursday, May 7 (lab):** analyze data in R, fit model, evaluate prediction quality.
- **Due Friday, May 15:** submit final deliverable to Canvas, one per team.
- After submission, Rosanna will release the actual answer for your prediction target.

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## Team structure and deliverables

You will work in teams of between 1 and 4 students for data collection and analysis. Each team will turn in:

1. **Team data file** (`.csv or link to google sheet`) with your measured variables
2. **Team analysis notebook** (`.ipynb`) that runs top to bottom.
3. **Team one-page PDF summary** (one page, 8.5 by 11 in.).

Your one-page summary should include:

- introduction stating the prediction problem,
- methods (data collection and data analysis),
- results (prediction equation, plugged-in prediction, and 95% prediction interval),
- conclusion in context,
- **team contributions section** listing what each member did and each team member's full name

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## Choose one project option

### Option A: Nesting doll prediction

Goal: predict the **heights of the inner dolls** only using measurements from the outer doll

Data source: your team will collect data from the 5 available doll sets in class. If you'd like, you may also use images of nesting dolls from the internet.

Examples of doll set variables:

- outermost doll height,
- outermost width,
- weight of doll set,
- sound of doll set when shaken,
- eye color of outermost doll, or

- any other measurable outer features

Be sure to collect data on the height of the inner dolls to create your prediction equation with! You may find it helpful to define **depth** as a variable, where depth 0 represents the outermost doll, depth 1 the next doll inside, and so on. Using the complete sets, you can record the height of each doll along with its depth and explore how height changes as depth increases. One possible approach is to model height as a linear function of depth and use this relationship to predict the heights of inner dolls for a new set when only the outermost height is known. If the outermost dolls vary in size across sets, you may instead consider working with the **proportion of the outermost height** at each depth (height at depth ÷ outer height), which can help account for differences in overall scale.

### Option B: Stone container prediction

Goal: predict the **number of stones in a container** given the total weight of the container and the stones.

Available tools include scales, rulers, and calipers.

Examples of stone set variables:

- total container weight,
- sample mean weight per stone from small random samples,
- sample variation in stone weights,
- optional size measures (e.g., sample diameters).

For the stone container problem, you may find it helpful to collect data on several samples with known numbers of stones and their corresponding total weights. This allows you to model the **number of stones as a function of total weight** using simple linear regression. In this setup, the slope represents how the number of stones changes with weight, and the intercept corresponds to the weight of the empty container. You can then use the given total weight of Rosanna's container as the predictor value to estimate the number of stones.

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## Suggested method: simple linear regression

You may wish to use **simple linear regression** as the prediction method:

```
lm(response ~ predictor, data = ...)
```

For this project:

- response = quantity to predict (height of doll,… or stone count),
- predictor = one chosen numerical or categorical predictor.

You may create and compare additional candidate predictors, but you should report at least one simple linear model. Your final prediction can be based on whatever method your team likes.

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## Data collection requirements

1. **Define variables before measuring.**
   - Write measurement rules so another team could replicate your process.
2. **Use consistent units** (for example: centimeters or grams).
3. **Double-check measurements** when feasible.
4. **Avoid convenience bias within your option** where possible (for example, sample stones from different parts of container rather than only top layer, don't avoid the fisherman nesting doll because it's inconveniently being used in a selfie with a local fisherman..).

## Data Recording

Record the data on a piece of paper and then tidy it up to share with your team for analysis in lab on Thursday. Remember our format for tidy data is to have the rows of a spreadsheet represent examples and the column the variables. It might be a good idea to have a team mate put the dataset into this format before your team is done with data collection so you can make sure you've collected all the appropriate information.

Once you have data recorded physically and it is tidy, you can add it to a spreadsheet, either Google Sheets or Excel. Be sure to give each column an informative name and make sure data on the same row is from the same doll or stone sample.

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## Data analysis workflow (lab on Thursday)

Your notebook should include the following sections.

### 1) Data loading and checking types

- Use `read_csv` from the `tidyverse` library or `read.csv` to load your spreadsheet. Remember to first save as .csv and then upload to Colab using the file manager on the left side bar.
- Use `glimpse` or `str` to check the type of each variable. You can use `as.numeric` or `as.factor` as needed to make each variable have the appropriate type in R.
- show first rows and variable names in R,
- show sample size (number of rows in the data frame in R)
- Both of these can be obtained by either `glimpse` or `str`. Remember that the spreadsheet in R becomes a `tibble` if you loaded it using `read_csv` from the `tidyverse` or a `data.frame` if you used `read.csv`.

### 2) Descriptive statistics and plots

- numerical summaries of response and predictor variables separately (be sure to include both a measure of center and spread). It is especially valuable to include the minimum and maximum values of the predictor variable as these will define what inputs are valid for your prediction model.
- numerical summary of the relationship between response and predictor (e.g. Pearson Correlation via `cor`)
- scatterplot with clear axis labels/units,
- brief statement on form, direction, and strength.

### 3) Linear regression fit

- fit `lm(response ~ predictor)`,
- Give the equation for the prediction, reporting the slope and intercept with interpretation in context. Be sure to include units.

### 4) Prediction

- generate one prediction with a 95% prediction interval for the project target. State what value of the predictor you plugged in from Rosanna's doll set or stone container.

## Format for one-page summary (required sections)

Use short clear headings:

1. **Introduction (problem statement).**
    - State which option you chose and exactly what quantity you are predicting and what information you are using to create that prediction.
2. **Methods - data collection.**

- Describe how measurements were collected in just enough detail that another team in class could read this section and collect data the same way that your team did.

3. **Methods - data analysis.**
   - Describe how you analyzed the data to build a prediction equation. (E.g. we used a scatterplot to assess linearity, used `lm` to build simple linear regression model of _____ on _____ and ___________ to obtain a 95% prediction interval at x=_______________).
4. **Results.**
   - Show the scatterplot with appropriate labels and title, commenting on form, direction and strength.
   - Report the value of Pearson's correlation from `cor`.
   - Write the prediction equation from the output of `lm`.
   - Briefly summarize when to use the prediction equation (range of valid inputs, units of both predictor and response variables)
5. **Conclusion.**
   - Plug in your measurement from the outer doll of Rosanna's doll set or the weight of Rosanna's container to your prediction equation and report your predictions of her inner doll heights or or her number of stones. Report an appropriate interval for the target prediction (use a prediction interval for an individual target value, `predict` with `interval="prediction"``).
6. **Team contributions (required).**
   - List each member's full name and their specific contributions (for example: data collection, measuring, idea generation, write-up, data analysis, data entry, organization).

Contribution section format example:

- Member A: collected measurements, and recorded them in a spreadsheet
- Member B: created plots and fit model.
- Member C: drafted the report.
- Member D: made initial suggestion for the predictor and edited final write-up.

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